Roider 2020 B-cell lymphoma — analysis summary

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Harmony (θ=10) integrates 9 patients despite a severe clonal batch effect: same-sample neighbour fraction 0.995→0.26; iLISI median 1.00→4.58 of 9. One connected component (1 speck removed). Mixed AND continuous.

Gate 2 — module geography → GO (non-branching)

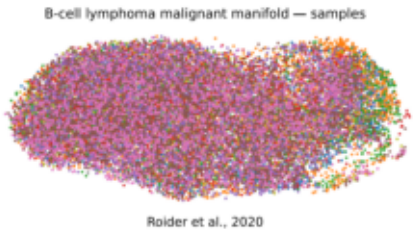
Non-branching case: a single (Gast/pSom)→Crest developmental gradient with organized Gast↔pSom cross-repression (smoothed r=−0.41; Crest↔pSom r=−0.52). Along pseudotime Gast/pSom fall (ρ −0.77/−0.15), Crest rises to a terminal tip (decile pt 0.77). Joint pSom&Crest enrichment 0.50 (<1 = inverse gradient, not a hub).

Gate 3 — cross-repression → NA

Not applicable — non-branching topology (principal curve, 2 tips / 0 forks, fold self-approach 2.95). No resolved bifurcation, so the branching-only hub cross-repression test is not run.

STEP FIGURES

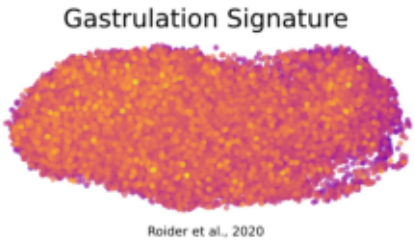
Manifold — samples



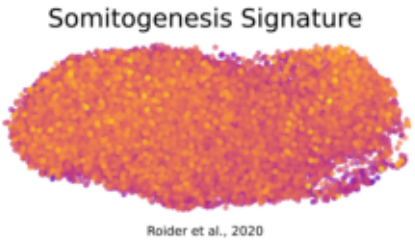
Pseudotime



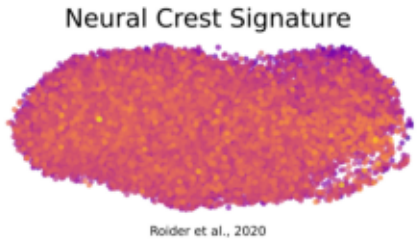
Gastrulation



Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
B-cell lymphoma lymph-node biopsies (Roider et al., 2020), 10x — 22,599 malignant cells over 9 samples (DLBCL/FL/tFL); reactive-lymph-node normals contribute no malignant cells; single low-cycling body (~11% cycling). Non-branching Linear-pSom/Crest.

Verdict.
Pattern reproduces in the non-branching form (Linear-pSom/Crest): one continuous gastrulation/somitogenesis → neural-crest axis, Crest terminal. G1 GO / G2 GO / G3 NA. EMT-TFs ZEB2 (early/root) → ZEB1 (terminus).

Parameter choices

Sample threshold	≥100 cells (largest gap 1583→327 above floor) → 9/9 samples	**Overdispersed genes**	1,334 (scFates find_overdispersed)
Integration	Harmony on sample θ=10 (mixing plateau; clonal patient batch)	**Embedding**	n_pcs=20, n_neighbors=30, spread=1.3; 1 island removed → 22,599
Geometry	non-branching curve N=20, λ=0.3, μ=100; 2 tips / 0 forks (2.95)	**Root**	tip 0 = Gast/pSom-high, Crest-low, low-cycling origin
Module means (raw)	Gast −0.027 · pSom +0.002 · Crest +0.005 (Linear-pSom/Crest)	**EMT-TFs**	ZEB2 (early/root) → ZEB1 (late/terminus); only ZEB1/ZEB2 clear >3% floor